

Turning Fragmented CCTV Into One Operational Intelligence Layer

The problem is not a lack of cameras. It is that ~80,000 cameras operate as fragmented, vendor-locked systems no human can continuously watch or manually search.

~26 Departments

Different Vendors

Different VMS / NVRs

Different Protocols

Geographically Dispersed Cameras

Manual Monitoring

Slow Investigation

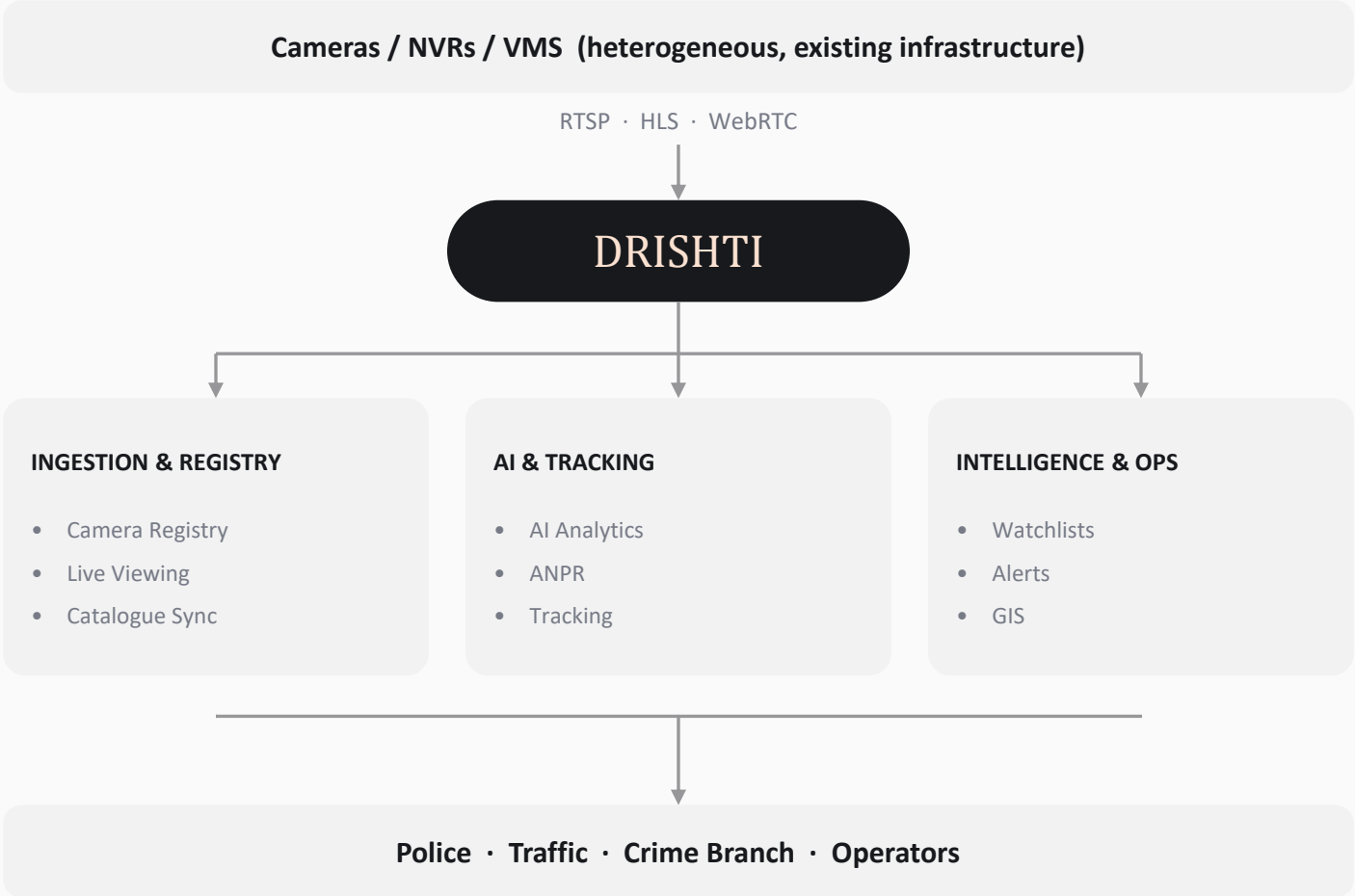
~80,000

CCTV cameras across Gujarat — the scale any solution must operate at.

CURRENT PAIN POINTS

- Vendor silos block a unified view
- Fragmented, manual monitoring
- Tedious post-incident investigation
- Disconnected data across departments
- Camera events cannot be correlated with watchlists
- All of the above — at statewide scale

One Intelligence Layer Above Existing CCTV



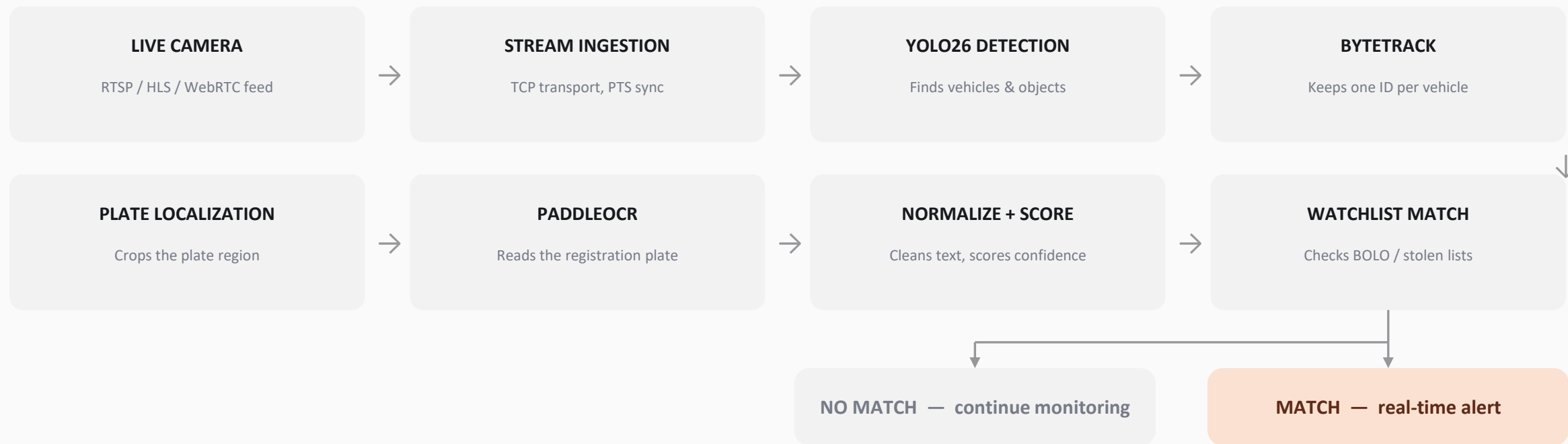
We do not replace existing cameras. We add an intelligence and integration layer above them.

KEY DIFFERENTIATORS

- IMPLEMENTED** Zero-leakage HLS proxy
- IMPLEMENTED** Track-caching ANPR
- IMPLEMENTED** Multi-camera journey correlation
- IMPLEMENTED** Immutable audit ledger

From Live Video to an Actionable Alert

IMPLEMENTED PIPELINE



This is the actual pipeline running in the Drishti PoC — not a conceptual diagram. A matched alert surfaces to the operator, GIS and analytics; a non-match keeps the vehicle under routine tracking.

Execution: runs on CUDA (NVIDIA GPU, FP16 via PyTorch) and gracefully degrades to AVX2 CPU when no GPU is available.

Detection Is Only the Beginning

AI-DERIVED

- Vehicle detection
- Person detection
- Object detection
- Multi-object tracking
- ANPR / OCR
- Plate normalization
- Confidence scoring

SYSTEM ANALYTICS

- Camera activity
- Event trends
- ANPR activity
- Camera status
- Department statistics
- Operational statistics
- Searchable vehicle sightings

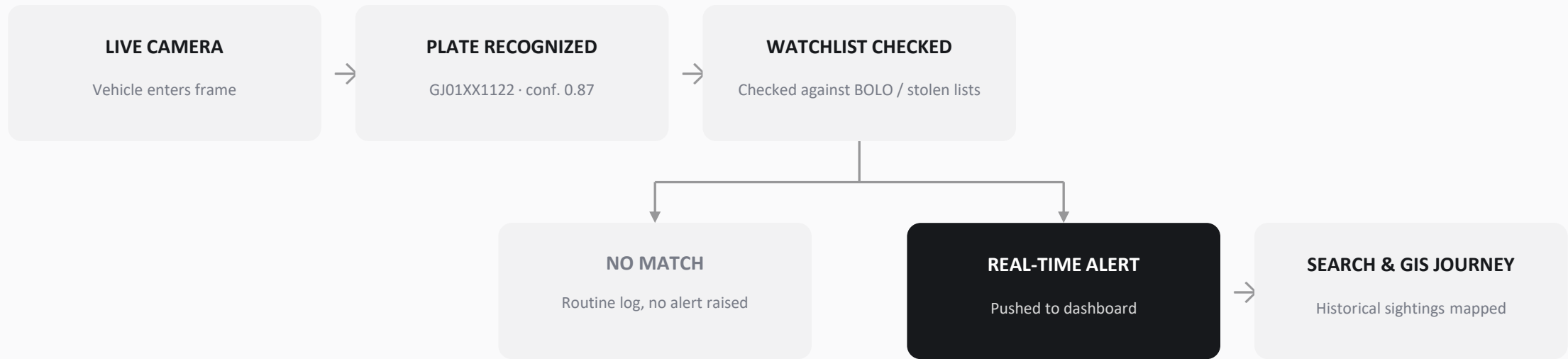
ACTIONABLE OUTPUT

- Real-time alerts
- Investigation workflow
- GIS journey correlation
- Evidence & event records
- Reports

TRACK-CACHING: THE KEY EFFICIENCY INNOVATION

Once a tracked vehicle receives a reliable plate read, repeated OCR is skipped for the rest of that track — PaddleOCR is computationally expensive, so this avoids redundant processing while tracking continues, especially at busy intersections.

From Detection to Investigation



Watchlist data used in the PoC is synthetic. Live VAHAN / eGujCop integration is proposed, not currently connected.

MANUAL

Search multiple camera feeds manually, cross-referencing footage by hand across departments and vendors.

DRISHTI

Search plate → retrieve historical sightings → visualize the vehicle's journey on the map.

Designed to Scale: Edge → Regional → Central

CENTRAL COMMAND — GANDHINAGAR

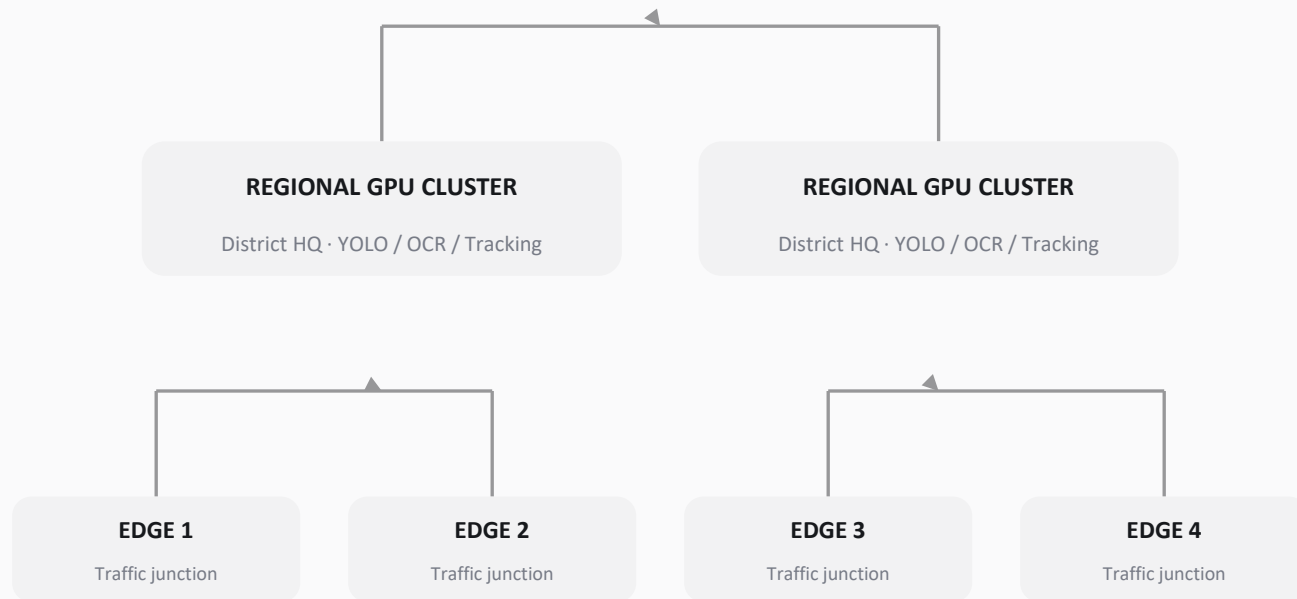
Watchlists · Alerts · GIS · Search · Analytics · APIs / Metadata

CURRENT PoC

Single-node, local processing on one workstation.

TARGET

Distributed statewide architecture across ~80,000 cameras.



Sending 80,000 HD video streams to a central server is impossible. Video stays close to the source — only structured metadata (plate, timestamp, camera, track/event, alerts) moves centrally.

PROPOSED TECHNOLOGIES

PROPOSED

Kubernetes

PROPOSED

NVIDIA Triton

PROPOSED

Kafka / RabbitMQ

PROPOSED

PostgreSQL scaling / ClickHouse

Technically Feasible. Economically Scalable.

HARDWARE

- PoC: single workstation, RTX 3060/4060, 16GB RAM
- Target: regional NVIDIA L4 GPU clusters

NETWORK

- Camera → edge/regional; metadata → central
- SD-WAN approach; avoids raw-video transport centrally

STORAGE

- Metadata retained centrally (PostgreSQL)
- Video on edge NVRs, ~30-day retention; evidence uploaded selectively to central object storage

PRODUCTION READINESS

- High availability
- Backup
- Disaster recovery
- Monitoring

COST MODEL

Camera Count × Video Bitrate × Retention × AI Processing Rate × GPU Capacity × Network Infrastructure

CAPEX: compute, GPU, networking, storage · OPEX: bandwidth, power, maintenance, storage, support

Final statewide cost requires department-level camera, bitrate, retention and infrastructure inventory.

One Platform. Statewide Visibility. Actionable Response.

SECURITY

IMPLEMENTED	JWT authentication
IMPLEMENTED	RBAC (Admin / Officer / Traffic)
IMPLEMENTED	Immutable audit logging
IMPLEMENTED	Server-side stream credential protection
IMPLEMENTED	Encrypted stream proxying
PROPOSED	mTLS between edge & central
PROPOSED	SSO / Active Directory integration
PROPOSED	Hardware Security Modules (HSM) for keys

IMPACT

Reactive	→	Proactive
Manual CCTV review	→	Searchable events
Fragmented feeds	→	Unified operational view
Manual plate search	→	Automated ANPR
Disconnected watchlists	→	Real-time correlation

DRISHTI GUJARAT

Transforming Gujarat's existing CCTV infrastructure into a unified operational intelligence platform.